3-1 Anaphylaxis v.1

- Unexplained hypotension
- Unexplained tachycardia or bradycardia
- Unexplained severe bronchospasm
- Angioedema
- Cardiac arrest where other causes are excluded
- Cutaneous flushing in association with one of more of the signs above

START

1. Call for help and inform theatre team of problem. Note the time.
2. Call for cardiac arrest trolley.
3. Remove all potential causative agents and maintain anaesthesia.
4. Give 100% oxygen and ensure adequate ventilation:
    - Maintain the airway and, if necessary, secure it with tracheal tube.
5. Elevate patient's legs if there is hypotension.
6. If indicated, start CPR immediately.
7. Give adrenaline (Box A):
    - Consider starting an i.v. infusion if repeat boluses required.
    - If blood pressure does not recover, consider alternate vasopressor (e.g. metaraminol).
    - Consider glucagon in ß-blocked patient unresponsive to adrenaline.
8. Give i.v. crystalloid at high infusion rate. (Adult 500-1000 ml; Child 20 $ml.kg^{-1}$).
9. If bronchospasm is persistent, consider → **3-4**
10. Take blood sample (5-10 ml clotted blood) for **serum tryptase** sample as soon as feasible.
11. Give secondary treatment drugs as soon as feasible (Box B).
12. Plan transfer of the patient to an appropriate critical care area.

Box A: ADRENALINE DOSES

- Adult: i.v. 50 µg (0.5 ml of 1:10 000 solution)
- Paediatric: i.v. 1.0 $µg.kg^{-1}$ (0.1 $ml.kg^{-1}$ of 1:**100 000** solution) [1:**100 000** solution made by diluting 1 ml 1:**10 000** up to 10 ml]
- If no i.v. access, intraosseous dose same as i.v.; adult i.m. dose 0.5mg

Box B: SECONDARY DRUGS

- Give chlorphenamine i.v.:
    - Adult: 10 mg
    - Child 6-12 years: 5 mg
    - Child 6 months-6 years: 2.5 mg
    - Child <6 months: 250 $µg.kg^{-1}$
- Give hydrocortisone i.v.:
    - Adult: 200 mg
    - Child 6-12 years: 100 mg
    - Child 6 months-6 years: 50 mg
    - Child <6 months: 25 mg

Box C: CRITICAL CHANGES

Cardiac arrest → **2-1**

Box D: OTHER REFERENCE INFORMATION

- Ensure subsequent repeat testing for serum tryptase.
- Liaise with hospital laboratory about timing and analysis of samples.
- Refer patient to a specialist allergy or immunology centre to identify the causative agent referred (Complete form and send to Dr Tomek).
- Inform the patient, surgeon and general practitioner.
- Report to MHRA (www.mhra.gov.uk/yellowcard).

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